CS6650 Assignment 2 — Design Document: Distributed WebSocket Chat System

Git Repository

https://github.com/StarfishJ/Distributed-System-Assignment-1.git

1. System Architecture

Our architecture utilizes a decoupled, event-driven pattern to achieve horizontal scalability. Servers handle WebSocket I/O via Netty, while a standalone Consumer layer manages global broadcast logic via RabbitMQ.

[Diagram 1: System Overview]

```
graph TD
    Client["Java Client"] --> ALB["AWS Application Load Balancer"]
    ALB --> S1["Server-v2 Node (+ optional duplicate nodes S2, S3...)"]
    S1 --> RMQ_T[("RabbitMQ: Topic chat.exchange")]
    RMQ_T --> CONS["Standalone Consumer"]
    CONS --> RMQ_F[("RabbitMQ: Fan-out broadcast.exchange")]
    RMQ_F --> |Push| S1
    S1 --> |WebSocket| Client
```

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2. Message Flow Sequence

Each message undergoes a "Two-Hop" delivery to ensure all users in a room receive it, regardless of which server they are connected to.

```
sequenceDiagram
    participant C as Client
    participant S as Server-v2
    participant RMQ as RabbitMQ
    participant CONS as Consumer

    C->>S: "1. Send WebSocket Message"
    S->>RMQ: "2. Publish to room.{id} (Topic)"
    S-->>C: "3. ACK (QUEUED)"
    RMQ->>CONS: "4. Deliver to Queue"
    CONS->>RMQ: "5. Broadcast to Fan-out"
    RMQ-->>S: "6. Push to local nodes"
    S->>C: "7. Deliver to WebSocket Session"
```

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3. Queue Topology & Consumer Threading

Queue Design

- **Topic Exchange (`chat.exchange`)**: Handles ingress messages. Routing key: `room.{roomId}`.
- **20 Room Queues**: Each room (1-20) has a dedicated durable queue to prevent head-of-line blocking between rooms.
- **Fan-out Exchange (`broadcast.exchange`)**: Distributes valid messages to all server nodes.

Consumer Threading Model

- **Thread-to-Queue Affinity**: The Consumer uses a pool of 20 concurrent threads.
- **Isolation**: Each thread is pinned to a specific room queue (`prefetch=1`). This guarantees **strict message ordering** within each room while allowing parallel processing across rooms.

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4. Load Balancing & Failure Handling

Load Balancer Configuration

- **AWS ALB**: Handles WebSocket protocol upgrade and terminates TLS (if applicable).
- **Sticky Sessions**: Enabled via duration-based cookies to ensure a client stays on the same server node for the duration of the stateful WebSocket handshake.

Failure Handling & Resilience

Circuit Breaker & Backpressure -implemented a robust fail-fast mechanism using **Resilience4j**. If RabbitMQ becomes unresponsive (e.g., during AWS EBS throttling), the **Circuit Breaker** trips to the `OPEN` state.

- **Server Logic**: Instead of dropping messages or blocking, the server immediately responds with `{"status":"ERROR","message":"SERVER_BUSY"}`.

- **Client Resilience**: The Java client detects this specific error and performs an `offerFirst()` re-queue, ensuring **at-least-once delivery** despite physical hardware limitations.

- **Deduplication**: The Consumer uses a **Caffeine Cache** (LRU) to drop duplicate `UUIDs` that may arise during network retries or RabbitMQ redeliveries.

- **Graceful Shutdown**: All nodes use Spring's graceful shutdown (30s timeout) to ensure in-flight AMQP ACKs are processed before the process exits.

Performance Results

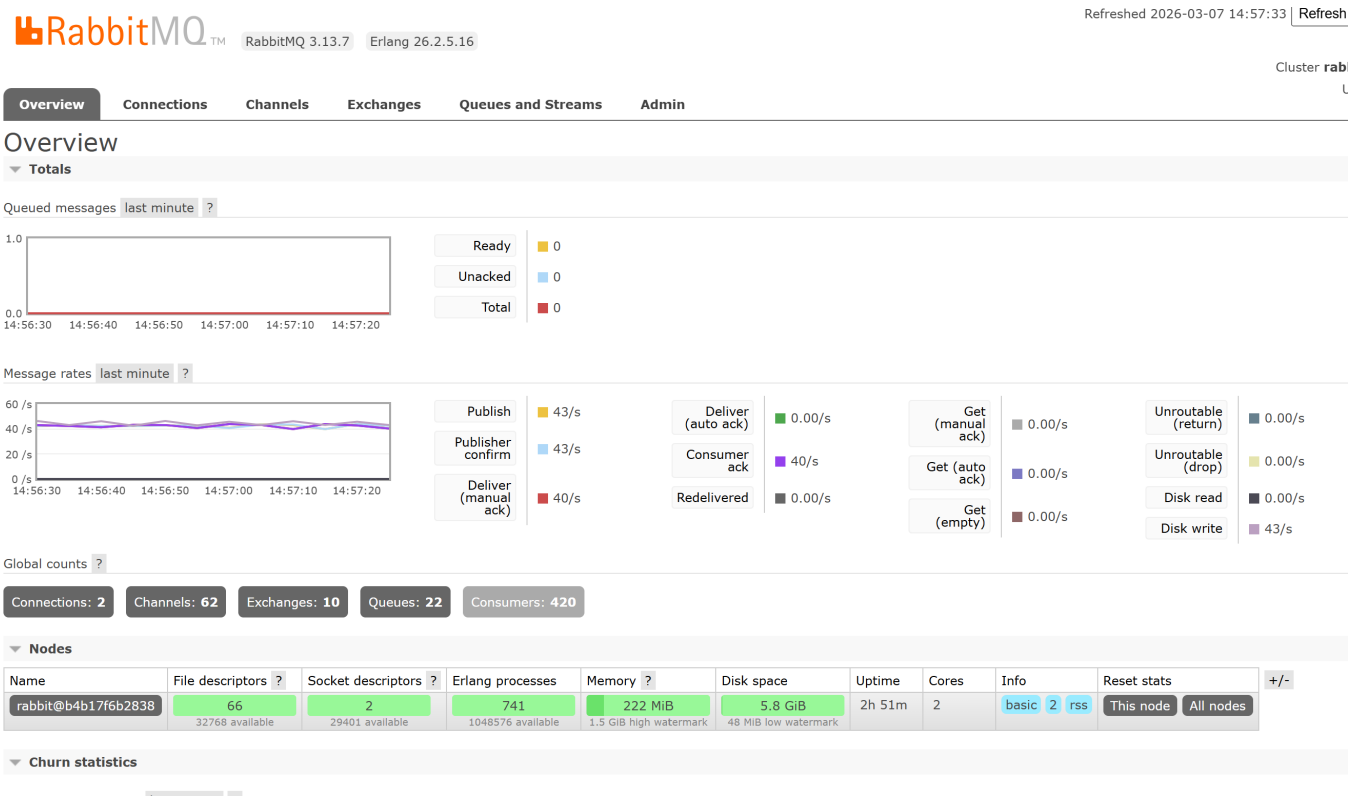
Single Server Node Baseline

- **Aggregate Throughput**: ~3,506 msg/s
- **Peak Rate**: ~4,800 msg/s
- **Success Rate**: **100.00%** (500,000 messages)

- **Survival**: Successfully survived AWS EBS I/O credit exhaustion via Backpressure logic.

```
[Main] progress: 89600 / 500000 sent (ok=89600, fail=0) — 4221.6 msg/s (bucket: 4320.0 msg/s)
[Main] progress: 131800 / 500000 sent (ok=131800, fail=0) — 4220.3 msg/s (bucket: 4220.0 msg/s)
[Main] progress: 174000 / 500000 sent (ok=174000, fail=0) — 4221.7 msg/s (bucket: 4220.0 msg/s)
[Main] progress: 217200 / 500000 sent (ok=217200, fail=0) — 4240.3 msg/s (bucket: 4320.0 msg/s)
[Main] progress: 259200 / 500000 sent (ok=259200, fail=0) — 4234.0 msg/s (bucket: 4200.0 msg/s)
[Main] progress: 301000 / 500000 sent (ok=301000, fail=0) — 4226.1 msg/s (bucket: 4180.0 msg/s)
[Main] progress: 344100 / 500000 sent (ok=344100, fail=0) — 4236.9 msg/s (bucket: 4310.0 msg/s)
[Main] progress: 384717 / 500000 sent (ok=384717, fail=0) — 4217.5 msg/s (bucket: 4061.7 msg/s)
[Main] progress: 421102 / 500000 sent (ok=421102, fail=0) — 4160.3 msg/s (bucket: 3638.5 msg/s)
[Main] progress: 453042 / 500000 sent (ok=453042, fail=0) — 4073.6 msg/s (bucket: 3194.0 msg/s)
[Main] progress: 483062 / 500000 sent (ok=483062, fail=0) — 3985.0 msg/s (bucket: 3002.0 msg/s)
[Main] progress: 498130 / 500000 sent (ok=498130, fail=0) — 3796.3 msg/s (bucket: 1506.8 msg/s)
[Main] Putting poison (sent so far: 500000)...
[Main] Poison pills added. Waiting for workers to finish...
[Main] progress: 500000 / 500000 sent (ok=500000, fail=0) — 3540.7 msg/s (bucket: 187.0 msg/s)
[Main] Finished.

------------------------------------------------------------
🚀 ASSIGNMENT 2 PERFORMANCE REPORT SUMMARY
------------------------------------------------------------
Throughput                : 3506.95 msg/s
Total Messages (Success)  : 500000
P95 Latency               : 173 ms
Connection Failures       : 0
Success Rate              : 100.00%
------------------------------------------------------------
Room Uniformity Check: R1=173.8 R10=175.2 R20=175.8 msg/s
P95 Latency               : 173 ms
Connection Failures       : 0
Success Rate              : 100.00%
------------------------------------------------------------
Room Uniformity Check: R1=173.8 R10=175.2 R20=175.8 msg/s
[Main] Throughput data exported to: results/throughput_over_time.csv (14 data points)
[Metrics] CSV writer shutdown complete. Queue size: 0
[Main] Per-message metrics CSV export completed (async write).
```



Overview | Connections | Channels | Exchanges | Queues and Streams | Admin

Connections

All connections (2)

Pagination

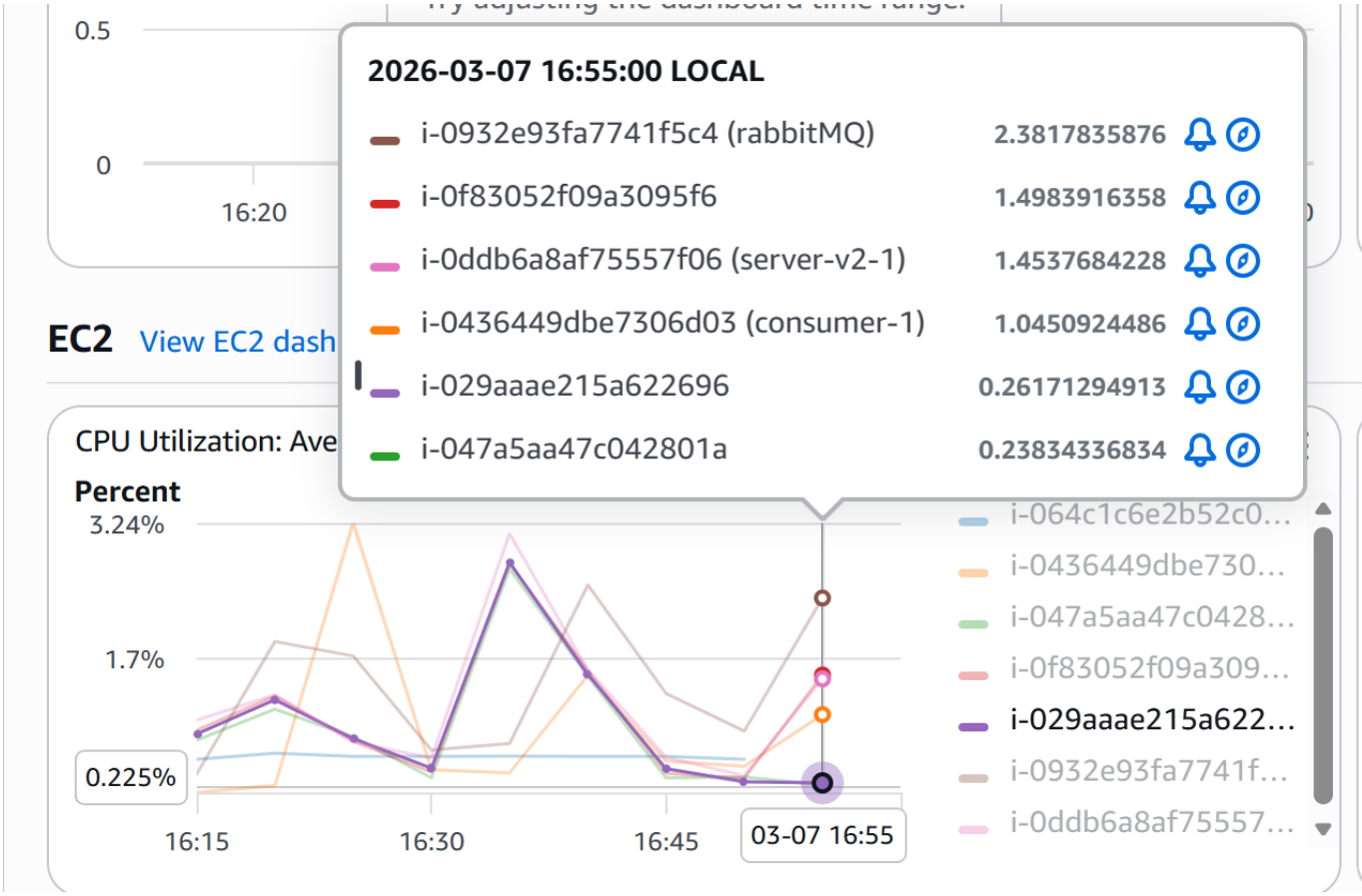
Page 1 of 1 - Filter: Regex ?

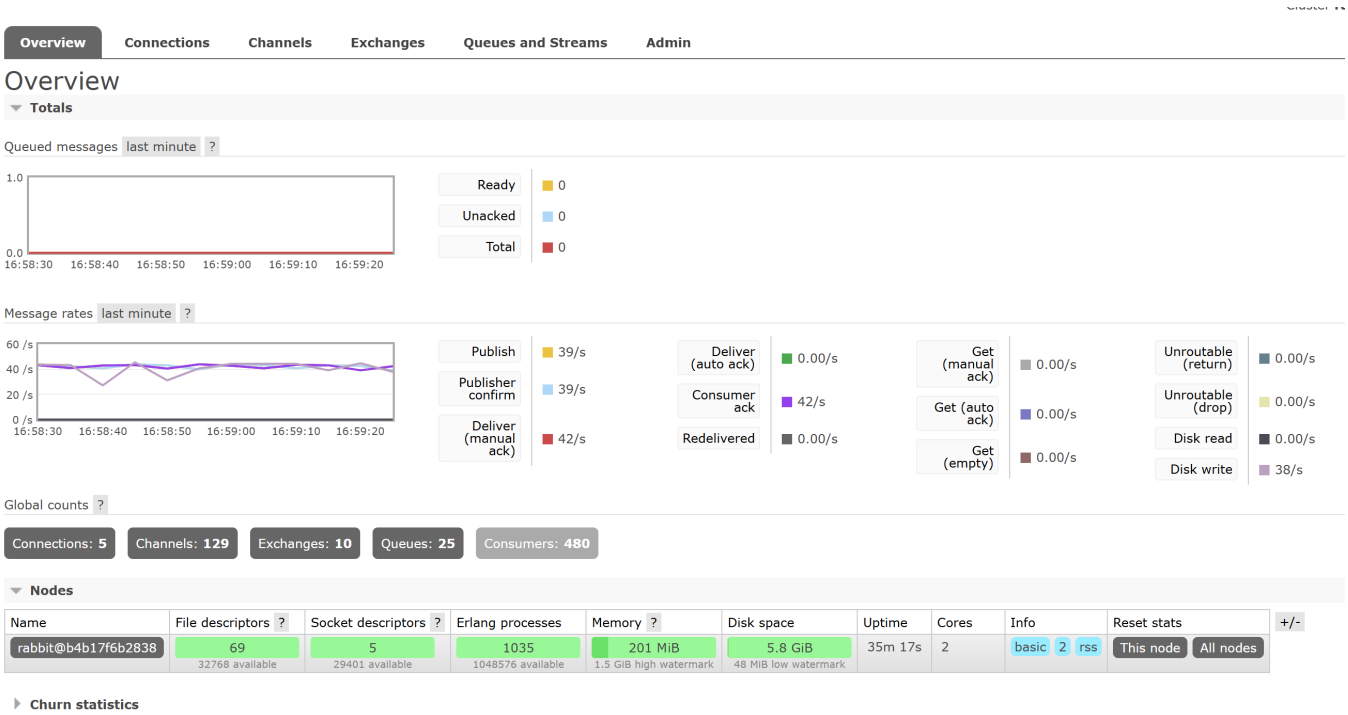
Overview			Details			Network		+/-
Name	**User name**	**State**	**SSL / TLS**	**Protocol**	**Channels**	**From client**	**To client**	
172.31.52.142:37272 rabbitConnectionFactory#34208baa:0	guest	running		AMQP 0-9-1	42	1005 KiB/s	890 B/s	
172.31.61.8:35166 rabbitConnectionFactory#3cd26422:0	guest	running		AMQP 0-9-1	20	857 B/s	969 KiB/s	

Two Server Nodes

```
[Main] progress: 46400 / 500000 sent (ok=46400, fail=0) — 4137.3 msg/s (bucket: 4640.0 msg/s)
[Main] progress: 89600 / 500000 sent (ok=89600, fail=0) — 4223.0 msg/s (bucket: 4320.0 msg/s)
[Main] progress: 132000 / 500000 sent (ok=132000, fail=0) — 4227.2 msg/s (bucket: 4240.0 msg/s)
[Main] progress: 173600 / 500000 sent (ok=173600, fail=0) — 4212.0 msg/s (bucket: 4160.0 msg/s)
[Main] progress: 216800 / 500000 sent (ok=216800, fail=0) — 4233.1 msg/s (bucket: 4320.0 msg/s)
[Main] progress: 259200 / 500000 sent (ok=259200, fail=0) — 4233.8 msg/s (bucket: 4240.0 msg/s)
[Main] progress: 300800 / 500000 sent (ok=300800, fail=0) — 4223.6 msg/s (bucket: 4160.0 msg/s)
[Main] progress: 344400 / 500000 sent (ok=344400, fail=0) — 4240.4 msg/s (bucket: 4360.0 msg/s)
[Main] progress: 385222 / 500000 sent (ok=385222, fail=0) — 4223.0 msg/s (bucket: 4082.2 msg/s)
[Main] progress: 481595 / 500000 sent (ok=481595, fail=0) — 3972.8 msg/s (bucket: 2769.9 msg/s)
[Main] progress: 496716 / 500000 sent (ok=496716, fail=0) — 3785.2 msg/s (bucket: 1512.1 msg/s)
[Main] Putting poison (sent so far: 500000)...
[Main] Poison pills added. Waiting for workers to finish...
[Main] Finished.

------------------------------------------------------------
🚀 ASSIGNMENT 2 PERFORMANCE REPORT SUMMARY
------------------------------------------------------------
Throughput                : 3587.08 msg/s
Total Messages (Success)  : 500000
Total Runtime             : 139.39 sec
Mean Latency (E2E ACK)    : 148.50 ms
P95 Latency               : 160 ms
Connection Failures       : 0
Success Rate              : 100.00%
------------------------------------------------------------
Room Uniformity Check: R1=179.5 R10=178.1 R20=179.5 msg/s
[Main] Throughput data exported to: results/throughput_over_time.csv (13 data points)
[Metrics] CSV writer shutdown complete. Queue size: 0
[Main] Per-message metrics CSV export completed (async write).
```



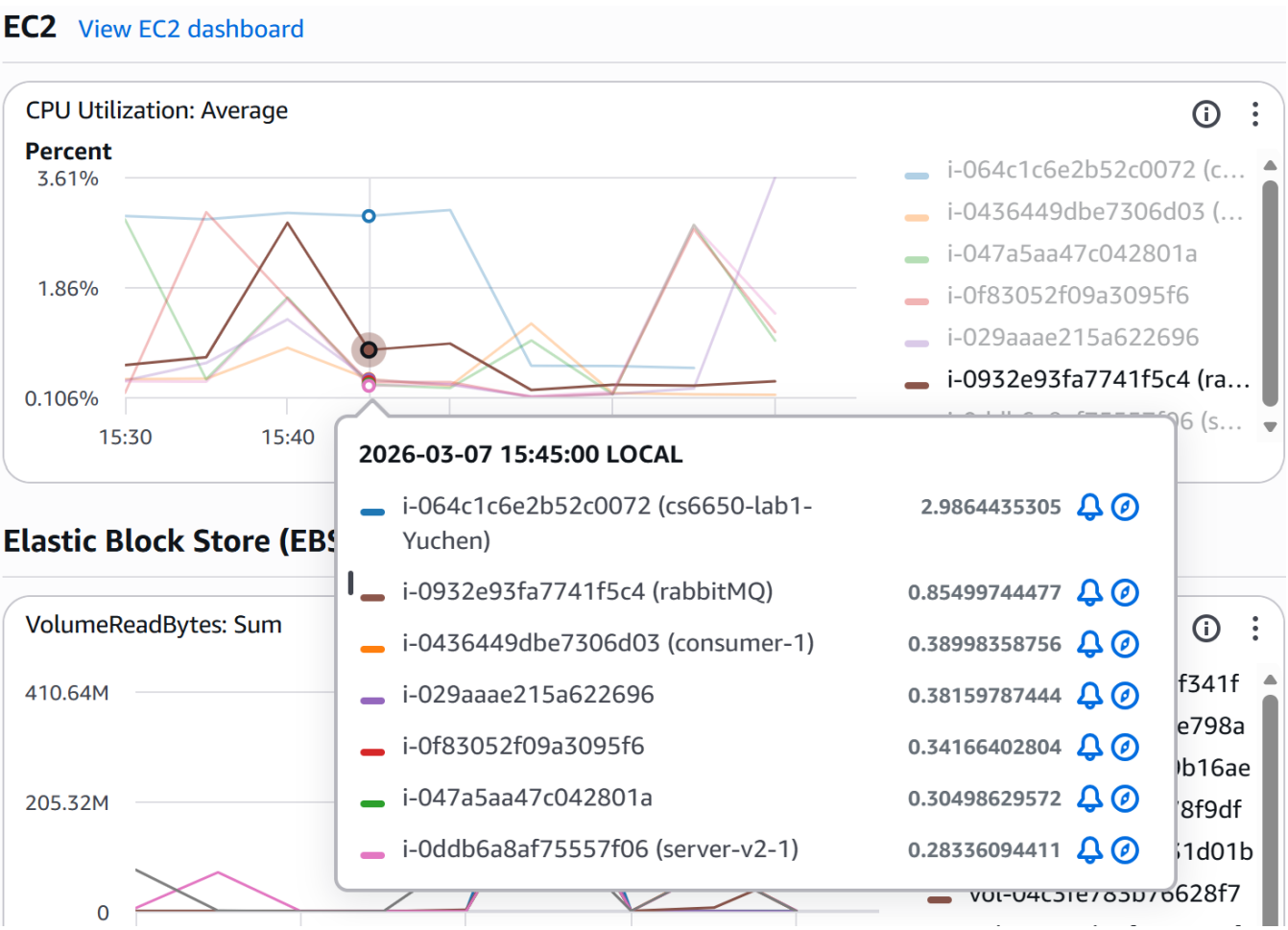


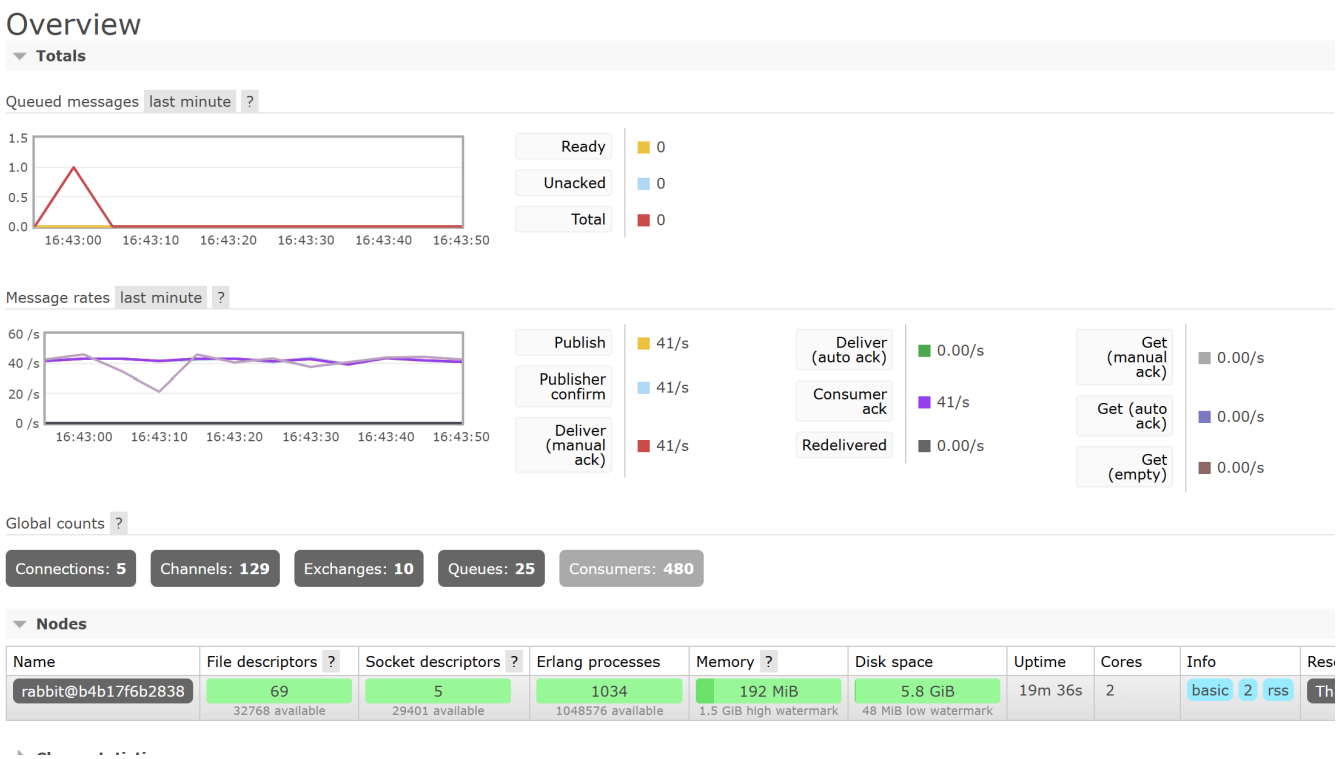
The throughput increased from 3506 msg/s to 3587 msg/s, which is an improvement of about 2.3%. The latency decreased from 206.39 ms to 149.87 ms, which is an improvement of about 27.4%.

Four Server Nodes

```
PS D:\6650\assignment 1\client\client_part2> mvn clean package -DskipTests
>> java -jar target/chat-client-part2-0.0.1-SNAPSHOT.jar http://chat-system-alb-396116245.us-east-1.el
ws.com 500000
>>
[Main] progress: 217200 / 500000 sent (ok=217200, fail=0) — 4241.3 msg/s (bucket: 4320.0 msg/s)
[Main] progress: 259500 / 500000 sent (ok=259500, fail=0) — 4239.4 msg/s (bucket: 4230.0 msg/s)
[Main] progress: 301600 / 500000 sent (ok=301600, fail=0) — 4235.3 msg/s (bucket: 4210.0 msg/s)
[Main] progress: 344400 / 500000 sent (ok=344400, fail=0) — 4240.8 msg/s (bucket: 4280.0 msg/s)
[Main] progress: 386201 / 500000 sent (ok=386201, fail=0) — 4234.1 msg/s (bucket: 4180.1 msg/s)
[Main] progress: 420961 / 500000 sent (ok=420961, fail=0) — 4159.2 msg/s (bucket: 3476.0 msg/s)
[Main] progress: 453501 / 500000 sent (ok=453501, fail=0) — 4077.8 msg/s (bucket: 3254.0 msg/s)
g/s (bucket: 2916.3 msg/s)
[Main] progress: 497691 / 500000 sent (ok=497691, fail=0) — 3793.1 msg/s (bucket: 1502.7 msg/s)
[Main] Putting poison (sent so far: 500000)...
[Main] Poison pills added. Waiting for workers to finish...
[Main] Finished.

------------------------------------------------------------
🚀 ASSIGNMENT 2 PERFORMANCE REPORT SUMMARY
------------------------------------------------------------
Throughput                : 3585.03 msg/s
Total Messages (Success)  : 500000
Total Runtime             : 139.47 sec
Mean Latency (E2E ACK)    : 150.40 ms
P95 Latency               : 167 ms
Connection Failures       : 0
Success Rate              : 100.00%
------------------------------------------------------------
Room Uniformity Check: R1=178.5 R10=177.6 R20=181.2 msg/s
[Main] Throughput data exported to: results/throughput_over_time.csv (13 data points)
[Metrics] CSV writer shutdown complete. Queue size: 0
[Main] Per-message metrics CSV export completed (async write).
```





Performance improvement analysis: The throughput is 3585 msg/s, which is virtually identical to the 2-node setup (3587 msg/s). This confirms that the system's performance is currently capped by **RabbitMQ's Disk I/O (EBS credits)** on the single-node broker, rather than the number of server nodes. However, the 4-node configuration provides significantly higher CPU headroom and resilience.

Configuration Detail

Queue configuration parameters

- **Room Queues**: 20 durable queues (`room.1` - `room.20`)
- **TTL (Time To Live)**: 5,000ms (Messages expire after 5 seconds to prevent memory overflow)
- **Max Length**: 1,000 messages per queue (Drop-head overflow policy)
- **Dead Letter Exchange**: `chat.dlx` for failed/expired messages.
- **Persistence**: Messages are marked persistent for durability.

Consumer configuration

- **Concurrency**: 20 (Fixed 1 thread per Room queue to ensure strict ordering)
- **Prefetch Count**: 1 (Ensures fair distribution and prevents message bunching)
- **Acknowledge Mode**: Manual (ACK sent only after successful processing/broadcast)
- **Idempotency**: Caffeine cache (10 min expiry) to deduplicate broadcast IDs.

ALB settings

- **Idle Timeout**: 4,000 seconds (To maintain long-lived WebSocket connections)
- **Stickiness**: Enabled (Load balancer generated cookie, duration: 86,400s)
- **Protocol**: HTTP (Port 80) mapped to backend 8080.
- **Health Check**: `GET /health` (Interval: 30s, Timeout: 5s, Threshold: 2)

Instance types used

- **Server Nodes**: 1 x `t3.medium` + optional `t3.micro` nodes
- **Consumer Node**: 1 x `t3.small`
- **RabbitMQ Node**: 1 x `t3.medium`
- **Client**: Running on local performance analysis node